

Operations Research

Linear Programming Models

Simplex method

Eg. 2: Maximize $Z = 23a + 32b$ subject to

$$10a + 6b \leq 2500$$

$$5a + 10b \leq 2000$$

$$1a + 2b \leq 500$$

Standardizing the above equation

$$\text{Max } Z = 23a + 32b - 0s_1 - 0s_2 - 0s_3$$

$$\text{Max } Z = 23a - 32b + 0s_1 + 0s_2 + 0s_3 = 0$$

$$10a + 6b + 1s_1 + 0s_2 + 0s_3 = 2500$$

$$5a + 10b + 0s_1 + 1s_2 + 0s_3 = 2000$$

$$1a + 2b + 0s_1 + 0s_2 + 1s_3 = 500$$

Table 1:

		Coefficients					RHS	RHS key column
		a	b	s_1	s_2	s_3		
R ₁	Z	-23	-32	0	0	0	0	$\frac{0}{-32} = 0$
R ₂	s_1	10	6	1	0	0	2500	$\frac{2500}{6} = 416.67$
R ₃	s_2	5 $\frac{1}{2}$	10 1	0	1 $\frac{1}{10}$	0	2000 200	$\frac{2000}{10} = 200$
R ₄	s_3	1	2	0	0	1	500	$\frac{500}{2} = 250$

The smallest positive ratio is 200

key row is R₃ and key element is 10

$$R_1 + R_3 (32) \quad -32 + 32 = 0 \quad [\text{key column}] \quad \boxed{2}$$

$$-23 + \frac{1}{2}(32) = -23 + 16 = -7$$

$$0 + 0(32) = 0 + 0 = 0$$

$$0 + \frac{1}{10}(32) = \frac{32}{10}$$

$$0 + 0(32) = 0$$

$$0 + 200(32) = 6400$$

Consider R_4 , $R_4 \rightarrow R_4 - R_3(2)$

$$2 - 2 = 0 \quad [\text{key column}]$$

$$1 - \frac{1}{2}(2) = 1 - 1 = 0$$

$$0 - 0(2) = 0$$

$$0 - \frac{1}{10}(2) = -\frac{1}{5}$$

$$1 - 0(2) = 1 - 0 = 1$$

$$500 - 200(2) = 100$$

In the Table 2, the Row R_1 has negative number

the maximum negative value is -7

The column corresponding to -7 will be key column

Circle the key column

Divide the RHS by key column

The minimum value is 185.75 which is R_2

the key row becomes R_2

Circle the key Row R_2

Table 2:

R_i	Basic variable	Coefficients					RHS	$\frac{\text{RHS}}{\text{Key column}}$
		a	b	s_1	s_2	s_3		
R_1	x	-7	0	0	$32/10$	0	6400	$\frac{6400}{-7} = -914.28$
R_2	s_1	$\frac{7}{1}$	0	1	$-\frac{3}{5}$	0	1300	$\frac{1300}{7} = 185.71$
R_3	s_2	$\frac{1}{2}$	1	0	$\frac{1}{10}$	0	200	$\frac{200}{\frac{1}{2}} = 400$
R_4	s_3	0	0	0	$-\frac{1}{5}$	1	100	$\frac{100}{0} = \infty$

we divide R_3 by 10 to get key element 1

$$\therefore R_3 \rightarrow R_3 \left(\frac{1}{10} \right) \quad [\text{Table 2 above}]$$

we have to make the key columns at R_2, R_1 & R_4 to zero.

Consider $R_2 \rightarrow R_2 - R_1(b)$ [Table 1]

$$6 - 1(b) = 0 \quad [\text{key column}]$$

$$10 - \frac{1}{2}(b) = 10 - 3 = 7$$

$$1 - 0(b) = 1 - 0 = 1$$

$$0 - \frac{1}{10}(b) = -\frac{3}{5}$$

$$0 - 0(b) = 0 - 0 = 0$$

$$2500 - 200(b) = 2500 - 1200 = 1300$$

Similarly, we make the key column at R_1 zero

$$R_1 \rightarrow R_1 + 32 R_3$$

In the Table 2, the key element is 7.

To make key element 1, Divide the R_2 by 7.

$$R_2 \rightarrow R_2 \left(\frac{1}{7} \right)$$

$$7 \left(\frac{1}{7} \right) = 1 \text{ [key element]}$$

$$0 \left(\frac{1}{7} \right) = 0$$

$$1 \left(\frac{1}{7} \right) = \frac{1}{7}$$

$$-\frac{3}{5} \left(\frac{1}{7} \right) = -\frac{3}{35}$$

$$0 \left(\frac{1}{7} \right) = 0$$

$$1300 \left(\frac{1}{7} \right) = \frac{1300}{7}$$

Table 3:

	Basic Variable	Coefficients					RHS
		a	b	s_1	s_2	s_3	
R_1	x_1	0	0	1	$13/5$	0	7700
R_2	s_1 a	<u>1*</u>	0	$1/7$	$-3/35$	0	$1300/7$
R_3	s_2	0	1	$-1/14$	$1/7$	0	$750/7$
R_4	s_3	0	0	0	$-1/5$	1	100

Consider Row R_1 , we need to make the key column zero

$$R_1 \rightarrow R_1 + R_2(7) \quad -7 + 1(7) = 0 \quad [\text{key column at } R_1]$$

$$0 + 0(7) = 0 + 0 = 0$$

$$0 + \frac{1}{7}(7) = 0 + 1 = 1$$

$$\frac{32}{10} + \frac{-3}{35}(7) = \frac{16}{5} - \frac{3}{5} = \frac{16-3}{5} = \frac{13}{5}$$

$$0 + 0(7) = 0 + 0 = 0$$

$$6400 + \frac{1300}{7}(7) = 7700$$

Consider Row R_3 , we need to make the key column at R_3 zero

$$\cancel{R_3 \rightarrow \frac{1}{2} R_2} \quad R_3 \rightarrow R_3 - R_2\left(\frac{1}{2}\right)$$

$$\frac{1}{2} - 1\left(\frac{1}{2}\right) = 0 \quad [\text{key column}]$$

$$1 - 0\left(\frac{1}{2}\right) = 1 - 0 = 1$$

$$0 - \frac{1}{7}\left(\frac{1}{2}\right) = -\frac{1}{14}$$

$$\frac{1}{10} - \left(\frac{-3}{35}\right)\left(\frac{1}{2}\right) = \frac{1}{10} + \frac{3}{70} = \frac{70+30}{700} = \frac{1}{7}$$

$$0 - 0\left(\frac{1}{2}\right) = 0 - 0 = 0$$

$$200 - \frac{1300}{7}\left(\frac{1}{2}\right) = 200 - \frac{650}{7} = \frac{750}{7}$$

Coming to Row R_4 , we notice that the key column is already zero.

$\therefore$ we leave the R_4 as it is

Consider Row R_1 , it is observed that the R_1 is positive.

We replace the slack s_1 on key row by the corresponding coefficient in key column
 s_1 is replaced ~~by~~ with a

The final value is $Z = 7700$

$$a = \frac{1300}{7} = 185.71$$

$b = ?$ (unknown)

From the objective function, $Z = 23a + 32b$

$$\therefore 7700 = 23(185.71) + 32b$$

$$7700 = 4271.33 + 32b$$

$$32b = 7700 - 4271.33$$

$$32b = 3428.67$$

$$b = \frac{3428.67}{32} = 107.14$$

$\therefore$ The answer is $Z = 7700$

$$a = 185.71$$

$$b = 107.14$$